



Lab task #03

Name: Zara Ahmed

Registration no: Fa25-BCS-060

Course name: Data Structure

Submission date: 21 September, 2026

Instructor: Bilal Bukhari

Code#01

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int var = 10;
```

```
    // declare pointer and store address of var
```

```
    int* ptr = &var;
```

```
    // print value and address
```

```
    cout << "Value of var: " << var << endl;
```

```
    cout << "Address of var: " << &var << endl;
```

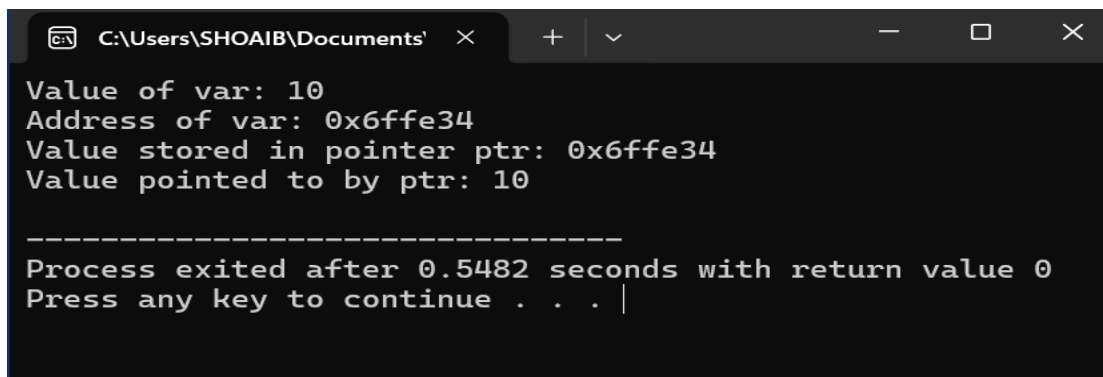
```
    cout << "Value stored in pointer ptr: " << ptr << endl;
```

```
    cout << "Value pointed to by ptr: " << *ptr << endl;
```

```
    return 0;
```

```
}
```

Output:



```
C:\Users\SHOAIB\Documents' x + v - □ x
Value of var: 10
Address of var: 0x6ffe34
Value stored in pointer ptr: 0x6ffe34
Value pointed to by ptr: 10
-----
Process exited after 0.5482 seconds with return value 0
Press any key to continue . . . |
```

Code#02

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int var = 10;
```

```
    // Store the address of
```

```
    // var variable
```

```
    int* ptr = &var;
```

```
    // Access value using (*)
```

```
    // operator
```

```
    cout << *ptr;
```

```
    return 0;
```

```
}
```

Output:

```
10
```

```
-----
```

```
Process exited after 0.8651 seconds with return value 0
```

```
Press any key to continue . . . |
```

Code#03

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int a = 10;
```

```
    int b = 99;
```

```
    int *ptr = &a;
```

```
    cout << *ptr << endl;
```

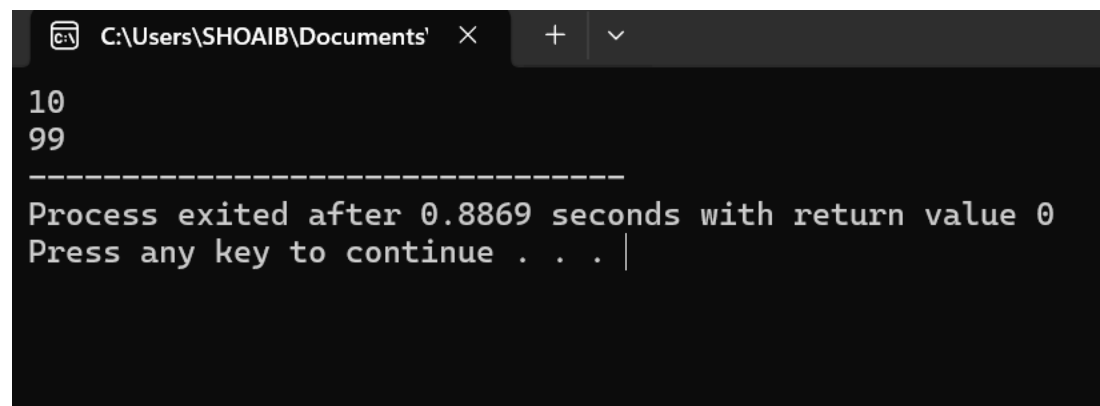
```
    // Changing the address stored
```

```
    ptr = &b;
```

```
    cout << *ptr;
```

```
    return 0;
```

```
}
```



The screenshot shows a Windows command prompt window with the title bar 'C:\Users\SHOAIB\Documents'. The output of the program is displayed in a monospaced font. It shows the value 10 on the first line, 99 on the second line, followed by a horizontal line of dashes. Below the dashes, it says 'Process exited after 0.8869 seconds with return value 0' and 'Press any key to continue . . . |'.

```
C:\Users\SHOAIB\Documents> 10
99
-----
Process exited after 0.8869 seconds with return value 0
Press any key to continue . . . |
```

Code#05

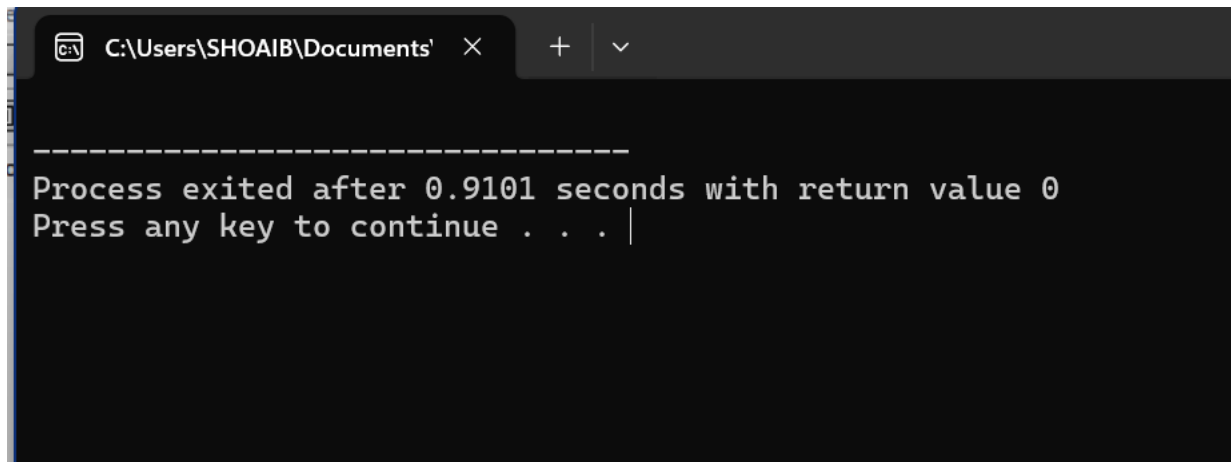
```
#include <iostream>

using namespace std;

int main() {

    // Wild pointer
    int *ptr;

    return 0;
}
```



```
-----
Process exited after 0.9101 seconds with return value 0
Press any key to continue . . . |
```

Code#06

```
#include <iostream>

using namespace std;

int main() {
```

```
// Wild pointer  
int *ptr;  
return 0;  
}
```

```
-----  
Process exited after 0.9462 seconds with return value 0  
Press any key to continue . . . |
```

Code#07

```
#include <iostream>  
  
using namespace std;  
  
int main() {  
    int nullptr;  
    // nullptr pointer  
    int *ptr = Null;  
  
    return 0;  
}
```

```
Pointer is currently null.
```

```
-----
```

```
Process exited after 0.8852 seconds with return value 0  
Press any key to continue . . .
```

Code#08

```
#include <iostream>
```

```
using namespace std;
```

```
int main() {
```

```
    int x = 42;
```

```
    // void pointer holding address of an int
```

```
    void* ptr = &x;
```

```
    // Error: cannot dereference void pointer
```

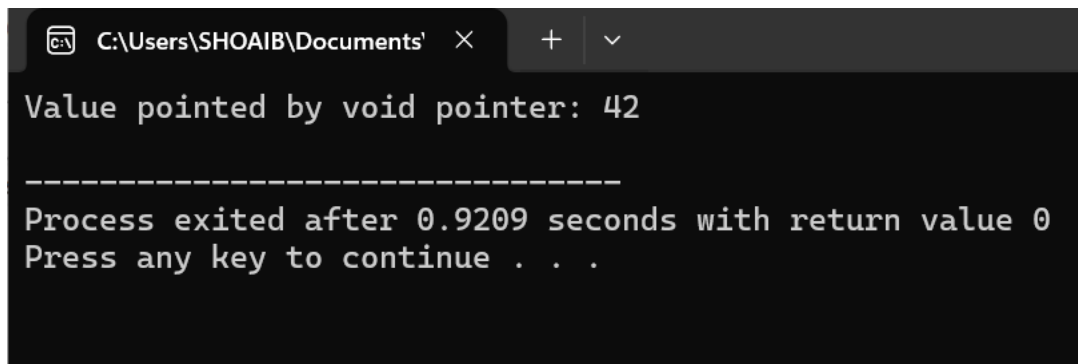
```
    // cout << *ptr;
```

```
    // Typecast before dereferencing
```

```
    cout << "Value pointed by void pointer: " << *(static_cast<int*>(ptr)) << endl;
```

```
    return 0;
```

```
}
```



```
C:\Users\SHOAIB\Documents' × + ∨  
Value pointed by void pointer: 42  
-----  
Process exited after 0.9209 seconds with return value 0  
Press any key to continue . . .
```

Code#10

```
#include <iostream>
```

```
using namespace std;
```

```
int* getPointer() {
```

```
    int x = 10;
```

```
    // returning address of local variable
```

```
    return &x;
```

```
}
```

```
int main() {
```

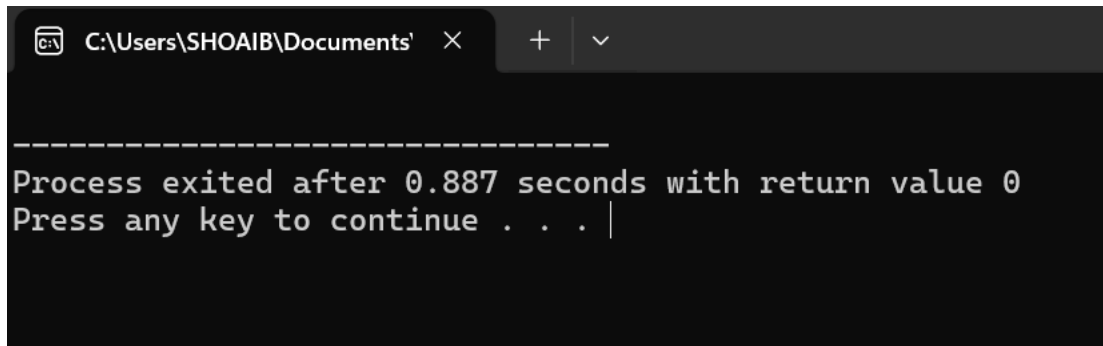
```
    // ptr becomes dangling here
```

```
    int* ptr = getPointer();
```

```
    // Undefined behavior
```

```
    // cout << *ptr;
```

```
    return 0;
}
```

A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\SHOAIB\Documents' and standard window controls. The command prompt displays the output of a program: a line of dashes, followed by 'Process exited after 0.887 seconds with return value 0', and then 'Press any key to continue . . . |' with a cursor at the end.

Code#11

```
#include <iostream>

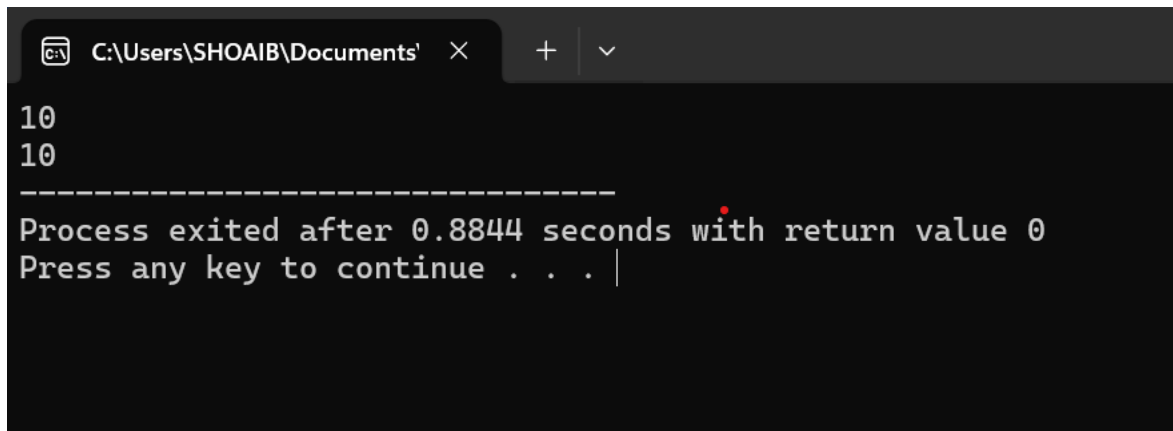
using namespace std;

int main() {
    int var = 10;

    // Store the address of
    // var variable
    int* ptr1 = &var;
    int** ptr2 = &ptr1;

    // Access values using (*)
    // operator
    cout << *ptr1 << endl;
    cout << **ptr2;
```

```
return 0;  
}
```



A screenshot of a Windows command prompt window. The title bar shows the file path 'C:\Users\SHOAIB\Documents' and standard window controls. The command prompt displays the output of a program: the number '10' is printed twice on separate lines, followed by a horizontal line of dashes. Below the dashes, a message states 'Process exited after 0.8844 seconds with return value 0'. The final line shows the prompt 'Press any key to continue . . . |' with a vertical cursor.

```
C:\Users\SHOAIB\Documents> 10  
10  
-----  
Process exited after 0.8844 seconds with return value 0  
Press any key to continue . . . |
```